BACSWN Mesh Architecture Master Plan

BACSWN Distributed Mesh Network — Master Architecture Plan

Document: BACSWN-MARCH-2026-002 **Classification:** CONFIDENTIAL — For Authorized Recipients Only **Prepared for:** Owner & CEO, BACSWN (private delivery partner) **Prepared by:** Sky Miles Limited — AI Elevate Division **Version:** 1.9 — June 2026 **Supersedes/extends:** BACSWN-ARCH-2026-001 (Distributed Intelligence Architecture Whitepaper) **Change log:** v1.9 adds the **trusted model supply-chain** constraint — no Chinese-built models (§4.1). v1.8 specifies the tower edge AI as a **multimodal model** (trained on weather + relevant data) running a **2-of-3 model-consensus ensemble** before autonomous action (§4.1), distinct from the multi-station consensus; notes the resulting Orin NX 16GB compute baseline. v1.7 adds **Annex A — RF Link Budgets** (VHF link budgets, radio-horizon analysis, and the engineering basis for the satellite-critical remote edge). v1.6 references the Master V&V and ICD Plan (BACSWN-VV-2026-006), which governs the per-station acceptance in §11 and the interface contracts across §4, §8, and §9. v1.5 adds **hub disaster recovery / control-plane redundancy** (§4.5) and **lightning, grounding & corrosion** hardening (§7.7); references the new O&M Plan (BACSWN-OM-2026-004) and Data Governance & Sovereignty Policy (BACSWN-GOV-2026-005). v1.4 names the primary radar as **Raytheon** equipment at designated radar towers (§8.8, cross-ref to Schedule WS-P) and adds **Tomorrow.io** space-based-radar weather ingestion (§4.3). v1.3 adds a sensing & response extensions section — distributed IoT sensor network and drone/UAS technology (§9). v1.2 extends §8 with commercial & military airspace surveillance (§8.8) and external/U.S. interoperability (§8.9). v1.1 added three mandated baselines — Category 4/5 hurricane survivability (§7), full end-to-end cryptographic security (§4.4), and emergency-services / inter-agency integration (§8) — with matching hardware, standards, and acceptance updates.

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1. Purpose & Scope

This Master Architecture Plan is the controlling engineering reference for the BACSWN distributed mesh network. Where the original whitepaper (BACSWN-ARCH-2026-001) made the *case* for moving from hub-and-spoke to an autonomous mesh, this document specifies the **buildable architecture** across two tightly coupled domains:

1. **Software architecture** — the edge-node software stack, the central hub/cloud services, the mesh protocol, the edge-AI runtime, data and security models, and the operator platform.
2. **Communications network architecture** — the physical and link-layer design for **each of the 15 radar/weather stations**, including primary, secondary, and tertiary transport, antennas, power, link budgets, and per-station resilience posture.

The plan covers all 15 stations of the Bahamas archipelago network and is the reference of record for procurement, site engineering, spectrum licensing, and acceptance testing.

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2. Architectural Principles

#	Principle	Implication
P1	**Autonomy first**	Every station must continue sensing, predicting, and alerting locally with zero connectivity to the Nassau hub.
P2	**No single point of failure**	The hub is a coordinator, not a dependency. Loss of Nassau degrades but never halts the network.
P3	**Defense in depth (comms)**	Each station carries three independent transport layers; failover is automatic and hysteresis-controlled.
P4	**Category 4/5 survivable**	The survival core of every node (edge compute, satellite link, battery, local alerting) is engineered to ride through a Cat 4/5 strike (≥157 mph / 252 km/h sustained) and keep warning. See §7.
P5	**Edge-correlated intelligence**	Hazard detection emerges from multi-station consensus, not a single sensor — eliminating false positives.
P6	**Open, versioned contracts**	Mesh protocol, message schema, and APIs are versioned; nodes of different firmware revisions interoperate.
P7	**Observable & serviceable**	Every node self-diagnoses sensor and link health and reports it; field service is data-driven.
P8	**Cryptographically secured end-to-end**	Every message, link, firmware artifact, and operator action is authenticated, integrity-protected, and (where applicable) encrypted from a hardware root of trust. No plaintext control path exists. See §4.4.

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3. System-Level Architecture

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│   NASSAU HUB / CLOUD CONTROL PLANE (MYNN)     │
│  • Operations dashboard (React SPA)           │
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                  │  • 7-agent orchestrator (wx-monitor, chief,    │
                  │    sigmet-drafter, flight-tracker, emissions,  │
                  │    qc, dispatch)                               │
                  │  • Hub AI models + forecast verification/MOS   │
                  │  • SQLite/WAL store, WebSocket fan-out         │
                  │  • Multi-channel dispatch (42 channels)        │
                  └────────────────▲───────────────▲───────────────┘
                                   │  (coordination, non-critical)
      VHF mesh backbone      ┌─────┴─────┐      Iridium SBD (always-on backup)
                             │           │
 ┌───────────┐      ┌───────────┐      ┌───────────┐             ┌───────────┐
 │ STATION   │◄────►│ STATION   │◄────►│ STATION   │◄── ... ──►│ STATION   │
 │ Edge Node │ VHF  │ Edge Node │ VHF  │ Edge Node │    VHF      │ Edge Node │
 └───────────┘      └───────────┘      └───────────┘             └───────────┘
 Each Edge Node = Autonomous Agent: local sensors + edge AI + mesh radio +
 satellite + cellular + solar/battery + local VHF/SMS alerting
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The network is a **partial mesh**: stations peer over licensed VHF where geography permits (see §6 topology), and every station additionally holds an always-on Iridium satellite path to the hub and to designated relay peers. The hub coordinates but does not gate station operation.

4. Software Architecture

4.1 Edge Node Software Stack (per station)

Layer	Component	Responsibility
L5 — Application	**Station Agent**	Local reasoning, microclimate prediction, anomaly/hazard detection, consensus participation, local alert decisions.
L4 — Edge AI Runtime	**Multimodal models + 2-of-3 ensemble**	Runs the on-node multimodal model (sensor time-series + radar/satellite imagery + METAR/text) for microclimate prediction, anomaly detection, and nowcasting; a **2-of-3 model-consensus ensemble** must agree before any autonomous correction/action. Trained on weather + site history; updated via signed OTA.
L3 — Mesh Protocol Stack	**Transport + routing**	Distance-vector routing optimized for low-bandwidth radio; store-and-forward; transport abstraction over VHF / cellular / satellite.
L2 — Data & State	**Local time-series + event store**	Ring-buffered observations, alert log, peer-link health, signed message journal for store-and-forward replay.
L1 — Sensor & Radio HAL	**Hardware abstraction**	Drivers for the sensor suite (temperature, pressure, wind, humidity, visibility, ceilometer) and the three radios.
L0 — Platform	**Hardened Linux on Jetson Orin**	Read-only root, watchdog, secure boot, OTA-updatable firmware with A/B rollback.

Multimodal edge model & on-node consensus. Each tower's decision model is **multimodal** — it fuses heterogeneous inputs (numeric sensor time-series, the Tomorrow.io space-based and Raytheon primary-radar imagery, and METAR/text) — and is **trained on weather and other relevant domain data** (site microclimate history, ERA5 normals). To make safe correction decisions, the node runs **three independent models** and requires a **2-of-3 majority** before acting — a triple-modular-redundancy / Byzantine-tolerant approach at the *inference* layer. This on-node model consensus is distinct from, and complementary to, the **multi-station** consensus in §4.2: one guards against a bad *model*, the other against a bad *station*. Running the multimodal ensemble drives the per-node compute baseline up to a Jetson **Orin NX 16GB** (with AGX Orin / independent triple-module options for higher assurance — see the Edge AI Node Hardware Budget, BACSWN-HWB-2026-009). Models are versioned and updated via signed OTA (§4.4.5). **Trusted model supply chain:** all models — base weights and fine-tunes — must come from **trusted, non-Chinese sources** (no Chinese-built model weights or training pipelines), with documented provenance and integrity verification, consistent with the system's sovereignty and US/allied interoperability posture (§4.4, §8.9; policy in BACSWN-GOV-2026-005).

4.2 Mesh Protocol

- **Message types:** `HEARTBEAT` (keepalive with sensor/battery/link health), `OBSERVATION`, `HAZARD`, `CONSENSUS_VOTE`, `PROPAGATION` (station-to-station forecast handoff), `ALERT`, `OTA_FRAGMENT`, `ROUTE_UPDATE`.
- **Routing:** simplified distance-vector tuned for 9.6 kbps radio links; metric blends hop count, measured link quality, and transport cost (VHF cheapest, satellite most expensive).
- **Store-and-forward:** messages are queued and replayed when a degraded peer/link recovers; every message carries a monotonic sequence and signature for de-duplication and integrity.
- **Consensus:** weather hazards are confirmed by a quorum of in-range stations before a network alert is raised, virtually eliminating single-sensor false positives.

4.3 Hub / Cloud Control Plane

Built on the **already-live BACSWN platform** (FastAPI + React, SQLite/WAL, WebSocket fan-out, APScheduler). Adds, for the mesh:

- **Mesh Coordinator service** — ingests `HEARTBEAT` / `OBSERVATION`, renders live topology, holds the authoritative-but-non-blocking network picture, and brokers OTA model/firmware rollouts.
- **Hub AI + Forecast Verification** — trains the bias-correction (MOS) models and ERA5 climate normals centrally, then pushes versioned model artifacts to the edge.
- **External weather data ingestion** — fuses public feeds (AWC, NWS, OpenMeteo) and the commercial **Tomorrow.io** feed, whose **space-based radar** adds precipitation coverage over open ocean where ground radar is sparse. External feeds are supplementary; BACSWN's own sensors/radar remain authoritative (no sole-source dependency).
- **Operator platform** — the existing dashboard gains a Mesh Network view (node health, link quality, consensus log, autonomous-mode indicators) already prototyped in `api/mesh.py`.

4.4 Cryptographic Security Architecture (full, end-to-end)

Security is a **mandated baseline**, not an add-on. The design provides confidentiality, integrity, authenticity, anti-replay, and non-repudiation across every plane — node-to-node, node-to-hub, firmware/model distribution, and operator access — anchored in a per-node hardware root of trust. **There is no unauthenticated or plaintext control path anywhere in the system.**

4.4.1 Hardware Root of Trust & Secure Boot

- Each node carries a **discrete secure element / TPM 2.0** (e.g., Microchip ATECC608B or equivalent FIPS-validated device) alongside the Jetson Orin security engine. Private keys are generated **inside** the secure element at provisioning and **never leave it**.
- **Measured secure boot:** signed bootloader → kernel → read-only rootfs, each stage verified before execution; boot measurements sealed to the TPM. **Anti-rollback fuses** prevent downgrade to a vulnerable firmware.
- **Tamper response:** enclosure intrusion or environmental tamper triggers **key zeroization**, so a physically captured or storm-scattered node yields no usable secrets.

4.4.2 Identity & PKI

- A **BACSWN national PKI**: an **offline root CA**, an online **issuing/intermediate CA**, and per-node **X.509 certificates** bound to the secure-element key. Each station, the hub, and each operator has a distinct cryptographic identity.
- **Certificate lifecycle:** automated enrollment at manufacture, scheduled rotation, and **revocation** via short-lived certs + signed revocation lists distributed over the mesh (offline-tolerant: nodes cache and honor the latest signed CRL without needing live OCSP).

4.4.3 Transport & Message Cryptography

- **IP transports (cellular / satellite-IP): mutual TLS 1.3** with certificate pinning.
- **Non-IP transports (VHF mesh, Iridium SBD):** because these are not IP-native, every payload is wrapped in a compact **authenticated-encryption envelope** sized for 9.6 kbps links:
  - **AES-256-GCM** (or AES-256-CCM) for confidentiality + integrity,
  - **Ed25519** signatures for sender authenticity and non-repudiation,
  - **monotonic sequence numbers + sliding replay window + timestamp** for anti-replay,
  - per-session symmetric keys derived via **ECDH (X25519) + HKDF**, rekeyed on a time/volume schedule for **forward secrecy**.
- Envelope overhead is minimized (truncated tags where risk-appropriate, header compression) so authenticated security is affordable on low-bandwidth radio.

4.4.4 Consensus & Mesh Integrity

- Every `CONSENSUS_VOTE`, `HAZARD`, and `ROUTE_UPDATE` is **signed**; the quorum logic rejects unsigned, duplicate, or out-of-window messages. This gives **Byzantine resistance** — a single spoofed or captured node cannot forge a network alert or poison routing.

4.4.5 Firmware / Model (OTA) Security

- All firmware and edge-AI model artifacts are **signed by the issuing CA** and verified by the secure element **before** A/B activation; anti-rollback enforced. A failed verification reverts to the last-known-good slot automatically.

4.4.6 Operator & API Plane

- Operator/API access is **JWT-gated** (already enforced and security-tested: **15/15 passing**), hardened with **MFA**, **role-based access control**, full **audit logging**, and centralized **secrets management** (no secrets in code/config; SecretRef indirection).

4.4.7 RF / Spectrum Security

- VHF frames carry cryptographic station identity + replay protection to resist **spoofing** on the shared licensed band. **Jamming resilience** is provided by frequency agility and automatic failover to the satellite layer, so denial of one medium never silences a station.

4.4.8 Crypto-Agility & Post-Quantum Path

- Algorithms are **negotiated and versioned**, not hard-wired, enabling rotation without re-flashing hardware. A defined migration path to **NIST post-quantum** primitives (**ML-KEM / FIPS 203** for key establishment, **ML-DSA / FIPS 204** for signatures) protects long-lived national infrastructure.

4.4.9 Standards Alignment

- **FIPS 140-3**-validated crypto modules, **NIST SP 800-53** controls, **IEC 62443** for the OT/industrial-control surface, and ICAO security guidance for aviation data.

4.5 Hub Resilience & Disaster Recovery

The mesh is engineered to survive without the hub (P1/P2) — but the **hub control plane itself** (Common Operating Picture, dashboard, coordination, external integrations) must also survive a direct strike on Nassau:

- **Active-standby hub:** a geographically separate secondary hub (cloud region and/or a hardened off-island site) holds continuously replicated state and the COP; **automatic failover** with a defined **RTO/RPO**.
- **Data replication & backup:** the SQLite/WAL store streams to the standby and to encrypted off-island backups; **point-in-time recovery**.
- **Degraded-mode operation:** if *both* hubs are unreachable, stations continue in autonomous mode (§5.4) and reconcile via store-and-forward on recovery — **no data loss**.
- **Failover drills** are part of program acceptance, alongside the per-station autonomous-mode drill.

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5. Communications Network Architecture

5.1 Three-Layer Transport Model

Layer	Technology	Typical Range	Bandwidth	Role	Resilience
Primary	Licensed VHF radio mesh (148–174 MHz)	100+ nm	9.6 kbps	Station-to-station mesh backbone, local broadcast	Survives extreme weather; line-of-sight + repeater
Secondary	Cellular (LTE/where coverage exists)	Island cell coverage	Mbps	High-rate hub sync, OTA model/firmware pull	Fails early in storms / power loss
Tertiary	LEO satellite (Iridium 9603 SBD)	Global	2.4 kbps	Always-on backup, remote-station lifeline	Hurricane-proof; antenna is the survivable element

Each station runs **all applicable layers concurrently** with automatic, hysteresis-controlled failover. Order of preference for hub sync: Cellular → VHF-relay → Satellite. Order for mesh/consensus: VHF → Satellite. Local alerting (VHF voice/SMS gateway) is independent of all backhaul and works in full isolation.

5.2 Per-Station Hardware Baseline (all 15 nodes)

Component	Specification
Compute	NVIDIA Jetson Orin Nano (8 GB) — 40 TOPS edge AI inference
Primary radio	Motorola MTR3000 VHF repeater (licensed 148–174 MHz) + omni/yagi antenna
Satellite	Iridium 9603 SBD transceiver, hurricane-rated patch antenna
Cellular	Industrial LTE modem with external antenna (coverage sites only)
Sensor suite	Temperature, pressure, wind, humidity, visibility, ceilometer
Power	48 V DC solar array (stowable/cyclone-rated) + battery backup + grid tie + automatic transfer
Security	Discrete secure element / TPM 2.0 (FIPS-validated) holding per-node keys; tamper-zeroization
Enclosure	NEMA-4X / IP66, large-missile-impact-rated (ASTM E1996/E1886), surge-elevated mounting
Structure	Mast/tower to TIA-222-H / ASCE 7 Risk Category IV, ≥180 mph design wind (see §7)

5.3 Per-Station Communications Profile

Tiers: **Hub** = national coordinator; **Major** = regional sub-hub / high traffic; **Regional** = standard node; **Remote** = isolated, satellite-critical. VHF# = count of stations reachable on the primary VHF mesh.

Code	Station	Tier	Cell	Sat	VHF#	Primary VHF neighbors (nm)	Resilience posture
MYNN	Nassau / Lynden Pindling Intl	Hub	✅	✅	5	MYBC(32), MYCB(53), MYEM(63), MYER(71)	Hub: redundant power + dual cellular; coordinates but non-blocking

Code	Station	Tier	Cell	Sat	VHF#	Primary VHF neighbors (nm)	Resilience posture
MYGF	Freeport / Grand Bahama Intl	Major	✅	✅	4	MYBS(60), MYAT(71), MYBC(81), MYAM(87)	Northern sub-hub; relays Abaco/Bimini cluster
MYAM	Marsh Harbour / Abaco	Major	✅	✅	5	MYAT(22), MYBC(78), MYEM(84), MYGF(87)	Abaco sub-hub; hurricane-history priority site
MYBC	Chub Cay / Berry Islands	Regional	—	✅	8	MYNN(32), MYBS(77), MYCB(77), MYAM(78)	**Central mesh relay** — best-connected node; satellite-backed (no cell)
MYAT	Treasure Cay / Abaco	Regional	✅	✅	3	MYAM(22), MYGF(71), MYBC(84)	Pairs with Marsh Harbour
MYEM	Governor's Harbour / Eleuthera	Regional	✅	✅	5	MYER(25), MYNN(63), MYAM(84), MYBC(84)	Eleuthera north anchor
MYER	Rock Sound / Eleuthera	Regional	✅	✅	5	MYEM(25), MYNN(71), MYEG(82), MYCB(89)	Eleuthera south anchor
MYCB	Congo Town / Andros	Regional	✅	✅	4	MYNN(53), MYBC(77), MYER(89), MYEM(96)	Western flank; links Nassau ↔ Berry ↔ Andros
MYBS	South Bimini	Regional	✅	✅	2	MYGF(60), MYBC(77)	Western gateway; thin VHF, cell-backed
MYLD	Deadman's Cay / Long Island	Regional	✅	✅	3	MYEG(49), MYES(62), MYRD(69)	**Southern chain anchor** — gateway to remote south
MYEG	Exuma International	Regional	✅	✅	4	MYLD(49), MYES(80), MYER(82), MYRD(83)	Central-south junction
MYES	San Salvador	Remote	✅	✅	2	MYLD(62), MYEG(80)	Eastern outlier; satellite-critical in storms
MYRD	Duncan Town / Ragged Island	Remote	—	✅	2	MYLD(69), MYEG(83)	No cell; VHF to southern chain + satellite
MYMM	Mayaguana	Remote	—	✅	1	MYIG(92)	**Far SE edge** — VHF only to Inagua; satellite is lifeline
MYIG	Inagua / Matthew Town	Remote	—	✅	1	MYMM(92)	**Far SE edge** — VHF only to Mayaguana; satellite is lifeline; first landfall sentinel

Key engineering consequences

- **Chub Cay (MYBC)** is the structural backbone of the mesh (8 neighbors). It must be engineered as a hardened relay — extended battery autonomy, repeater-grade VHF, redundant satellite — even though it is a small island with no cellular.
- **Inagua (MYIG) and Mayaguana (MYMM)** are the network's remote edge: each has exactly one VHF peer (each other, at 92 nm) and **no cellular**. Their Iridium satellite path is mission-critical and must be the *first* layer commissioned. Inagua is also the southernmost sentinel — typically the Bahamas' first contact with an incoming Atlantic hurricane — so it carries outsized early-warning value.
- The **southern chain** (MYRD ↔ MYLD ↔ MYEG ↔ MYES) is a thin, near-linear topology. Long Island (MYLD) is the single articulation point connecting the remote south to the rest of the mesh; losing it isolates Ragged Island and weakens the path to San Salvador. It warrants relay-grade hardening and a guaranteed satellite fallback.

5.4 Failover & Autonomous Operation

1. **Normal:** cellular for hub sync, VHF for mesh/consensus, satellite idle (heartbeat only).
2. **Cellular loss (storm onset):** hub sync migrates to VHF-relay → satellite; mesh continues on VHF.
3. **VHF degradation (mast/antenna damage):** affected node falls back to satellite for both hub and critical peer messaging; neighbors re-route around it (distance-vector reconvergence).
4. **Hub unreachable (Nassau isolated):** stations enter **autonomous mode** — local edge AI keeps predicting; consensus and alerting continue among reachable peers; local VHF voice/SMS broadcast to aviation and emergency frequencies continues uninterrupted.
5. **Recovery:** store-and-forward replays the signed message journal; the hub reconciles and re-establishes the authoritative picture.

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6. Mesh Topology (derived)

Primary VHF adjacency (≤100 nm), forming three natural clusters bridged by Nassau and Chub Cay:

- **Northern cluster:** MYGF ↔ MYAT ↔ MYAM ↔ MYBS ↔ MYBC
- **Central cluster:** MYNN ↔ MYBC ↔ MYCB ↔ MYEM ↔ MYER
- **Southern chain:** MYER/MYEG ↔ MYLD ↔ MYRD / MYES, with **MYIG** ↔ **MYMM** as the detached SE pair bridged to the rest only via satellite.

Satellite (Iridium) overlays a **full logical mesh** on top of the VHF partial mesh, guaranteeing every station can always reach the hub and a designated relay regardless of VHF reachability.

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7. Hurricane Survivability — Category 4/5 Hardening

Design basis: every node must survive and keep warning through a **Saffir-Simpson Category 4/5 hurricane** — sustained winds ≥157 mph (252 km/h), gusts beyond 200 mph, multi-day grid and access loss, flying debris, and storm surge. Hurricane Dorian (2019) — which stalled over Abaco as a Cat 5 with ~185 mph winds and ~20 ft surge — is the **reference design event**.

7.1 Survival Core vs. Graceful Degradation

Components are tiered so that even total loss of the exposed elements never silences a station:

Tier	Components	Requirement under Cat 4/5
Survival core (must operate)	Edge compute, secure element, battery, **Iridium satellite** link, local VHF/SMS alert broadcast	Continues sensing, predicting, and warning throughout the storm
Resilient (may degrade, must recover)	VHF mesh radio + mast, solar array	May lose the mast/array at peak; node falls back to satellite; auto-recovers post-storm
Expendable (expected to fail)	Cellular, grid power, exposed non-critical sensors	Loss is planned-for and routed around

7.2 Structural & Wind Loading

- Masts/towers engineered to **TIA-222-H / ASCE 7** at **Risk Category IV** (essential facility), **≥180 mph design wind** with safety margin; foundations and guying certified by a licensed structural engineer per site survey.
- The **Iridium patch antenna is the designated survivable element** — low-profile, flush-mounted, cyclone-rated — chosen precisely because it presents minimal wind area. VHF masts use controlled-failure/breakaway design so a mast loss does not damage the survival core.

7.3 Debris, Water & Surge

- Enclosures **NEMA-4X / IP66** and **large-missile-impact-rated** (ASTM E1996/E1886) against flying debris.
- Equipment **elevated above base-flood + design surge** (site-specific; remote low-lying sites such as Inagua/Matthew Town and Ragged Island elevated to Dorian-class surge); sealed conduit and desiccant-managed enclosures against water ingress and humidity.

7.4 Power Autonomy

- 48 V solar + battery + grid tie with automatic transfer at every node. Because grid and physical access can be lost for **days to weeks**, batteries are sized for **extended autonomy**: standard nodes ≥72 h; **remote/relay nodes (MYBC, MYIG, MYMM, MYRD) target 96–120 h** with solar arrays that are **stowable or rated to survive** peak winds rather than acting as sails.

7.5 Communications Survivability

- The **satellite layer is the storm lifeline** — it is the one path engineered to remain up when VHF masts and cellular are gone, so the southern satellite-critical edge (Inagua, Mayaguana, Ragged Island) keeps reporting through landfall.
- Autonomous mode (see §5.4) means a node cut off from the hub still runs local edge AI and broadcasts local alerts on aviation/emergency VHF — exactly when it matters most.

7.6 Deployment Sequencing & Validation

- Stations are commissioned **south-to-north** so the highest-exposure, satellite-critical sites have a verified satellite lifeline **before June 1** hurricane season (Master Framework Schedule, gates G5/G9, risk R01).
- **Validation:** structural wind certification, missile-impact testing, surge-elevation sign-off, and an **extended-autonomy power-cut drill** are part of per-station acceptance (§11).

7.7 Lightning, Grounding & Corrosion

Tropical, marine, tall-mast sites face three constant threats beyond wind and surge:

- **Lightning protection:** air terminals/finials on every mast, down-conductors, and a low-impedance earth per **IEC 62305 / NFPA 780**; all radios and the edge node sit behind multi-stage **surge protective devices (SPDs)** on power, antenna, and data lines. Lightning is a *daily* Bahamian risk, not a storm-only one.
- **Grounding & bonding:** a single-point ground with measured low earth resistance; all metalwork, enclosures, and cable shields bonded to prevent ground loops and protect the secure element / Jetson from transients.
- **Salt-fog corrosion:** marine-grade throughout — stainless / hot-dip-galvanized hardware, marine-rated connectors, conformal-coated electronics, gasketed desiccant-managed enclosures — to corrosivity category **ISO 9223 C5-M**. Corrosion is the slow killer of unattended coastal sites.

These are gated at design (STR-1) and verified at per-station acceptance (§11).

8. Emergency-Services & Inter-Agency Integration

BACSWN is designed for **full operational integration with the Bahamian emergency-response ecosystem** — not merely to publish weather, but to drive and coordinate the national response. Integration is **bidirectional**: outbound (alerting, evacuation directives, tasking) and inbound (agency resource status, road/port closures, field incident reports) into one shared picture. This builds directly on the platform's existing emergency, evacuation, and 42-channel dispatch services.

8.1 Integrated Agencies & Systems

Agency / System	Role in response	Integration
NEMA (National Emergency Management Agency)	National coordination, declarations, activation	Common Operating Picture + tiered activation triggers
Royal Bahamas Police Force	Public safety, evacuation enforcement, road closures	COP feed; closure/incident reports inbound; CAP alerts to dispatch
Bahamas Fire Services / Rescue	Fire, rescue, hazmat	Tasking via CAP/EDXL; field status inbound
Royal Bahamas Defence Force (Coast Guard)	Maritime SAR, vessel picture, outer-island support	Shared maritime/airspace picture; SAR handoff; pre-positioning
Emergency shelter system	Shelter activation, capacity, evacuee intake	Bidirectional: activation/occupancy in; surge-aware routing out
EMS / Ministry of Health	Medical response, casualty management	Alerts + COP; facility status inbound
Supporting: Port Department, Bahamas Power & Light, Dept. of Meteorology	Ports, power, official forecasts	Status exchange and advisory coordination

8.2 Common Operating Picture (COP)

A single shared, real-time situational picture — storm track/cone, surge & flood overlays, evacuation zones, shelter status, live flight & maritime tracks, and field incident reports — available to every agency through the BACSWN dashboard and via API to agency command centers. **One authoritative source of truth** replaces siloed, phone-and-fax coordination.

8.3 Alerting & Tasking (outbound)

- **Standards-based exchange: Common Alerting Protocol (CAP, OASIS)** for public and agency alerts; **EDXL** for resource and situation exchange — so existing agency CAD/dispatch systems can ingest BACSWN alerts natively.
- **Redundant delivery:** the existing 42-channel dispatcher reaches agency operations centers and field units over data + SMS + VHF + satellite simultaneously.
- **Tiered activation:** hurricane-category thresholds trigger pre-agreed, per-agency playbooks.

8.4 Shelter & Evacuation Integration

- Bidirectional link to the emergency shelter system: shelter **activation, real-time capacity/occupancy, and accessibility** feed BACSWN; the **evacuation engine** computes and pushes **surge-aware routing** from at-risk zones to the nearest open shelter (extends the platform's existing evacuation service).

8.5 Maritime & Coast Guard (RBDF)

- Shares the maritime/airspace picture for **search-and-rescue coordination** and outer-island logistics; hands off vessel and aircraft tracks; supports asset **pre-positioning** before landfall.

8.6 Interoperability Standards & Incident Command

- **CAP / EDXL** data exchange; alignment with the **Incident Command System (ICS)** so BACSWN outputs map cleanly onto the national command structure; **open APIs + webhooks** for agency CAD/COP systems.

8.7 Security, Sovereignty & Resilience of Integrations

- All inter-agency links inherit the **§4.4 cryptographic architecture** (mutual TLS, signed envelopes, RBAC, full audit). Data crossing into police/defence networks honors the BACSWN **data-sovereignty & governance policy** (national ownership; least-privilege, need-to-know sharing).
- **Resilient by design:** when terrestrial networks fail in a Cat 4/5, alerts and shelter/evacuation directives still reach agencies and shelters over the surviving **VHF + satellite** layers and each station's **local broadcast** — exactly when inter-agency coordination matters most.

8.8 Airspace Surveillance — Commercial & Military Overflights

BACSWN maintains a complete real-time picture of **all aircraft transiting the Bahamas FIR**, both commercial and military:

- **Cooperative traffic:** ADS-B / Mode-S/C surveillance (the platform's existing flight-tracker) — identity, position, altitude, and type for commercial and most general-aviation traffic.
- **Military & non-cooperative traffic:** military aircraft frequently suppress ADS-B or use encrypted **Mode 5**; complete coverage therefore requires **fusing primary/secondary surveillance radar and partner data-sharing feeds** (see §8.9) into the track picture. Primary radar is provided by **Raytheon radar equipment installed at designated radar towers** (a distinct, long-lead hardware/civil program — see Master Framework Schedule, workstream WS-P). The surveillance layer is designed as **pluggable**, so radar and military feeds can be fused without changing the application.
- **One feed, three outputs:** every tracked overflight simultaneously drives **(1) safety & airspace management**, **(2) overflight & landing-fee accounting** — the commercial revenue basis, since you can only bill what you can count — and **(3) CORSIA carbon-emissions calculation** (per-flight CO_2, already implemented). Surveillance is thus the shared backbone of both the safety mission and the BOO revenue model.

8.9 External & International Interoperability (incl. United States)

The Bahamas FIR abuts U.S.-controlled oceanic airspace; the network is designed to interoperate beyond national borders as a **phased, future capability** that does not gate the core deployment:

- **Aviation weather exchange:** ICAO **OPMET / WAFS** and **AFTN/AMHS**, so BACSWN SIGMETs and observations flow into the international system.
- **United States (anticipated): NOAA / NWS / National Hurricane Center** (forecast & storm data); the **FAA** — **Miami Oceanic (ZMA ARTCC)** and the **Miami WAFC** — for cross-border traffic and airspace coordination; and, where bilaterally agreed, **U.S. military / NORAD** channels for shared military-track awareness.
- **Sovereignty-preserving:** all external sharing is **least-privilege, need-to-know**, governed by the data-sovereignty & governance policy and secured by the §4.4 cryptographic architecture. The Bahamas remains the data owner; cross-border feeds are bilateral and revocable.

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9. Sensing & Response Extensions — IoT Sensor Network & Drones

Beyond the 15 fixed stations, two extension layers densify sensing and add an **active** response capability. Both attach to the existing mesh, Common Operating Picture, and §4.4 security model.

9.1 Distributed IoT Sensor Network

- **Purpose:** low-cost densification of the fixed-station grid — micro weather stations, tipping rain gauges, anemometers, **storm-surge / tide / flood-level sensors**, lightning detectors, soil-moisture, and marine/buoy sensors — for settlement-level microclimate and flood awareness.
- **Connectivity:** low-power wide-area (**LoRaWAN / NB-IoT**), with each BACSWN station acting as the **LoRaWAN gateway** for its island cluster; readings aggregate up the mesh to the COP. Sensors run for years on solar/battery.
- **Edge correlation:** the station's edge AI fuses dense IoT readings into its microclimate and surge/flood nowcasts; multi-sensor consensus suppresses false readings.
- **Security:** constrained-device profile — **DTLS / lightweight AEAD**, per-device keys, signed payloads — inheriting the §4.4 PKI; spoofed or compromised sensors are quarantined.

9.2 Drone (UAS) Technology

- **Post-storm damage assessment:** rapid aerial survey of cut-off settlements when roads/ports are out, feeding imagery into the COP for NEMA and responders.
- **Weather reconnaissance:** small UAS for atmospheric profiling and sampling the storm environment where fixed sensors cannot reach.
- **Search-and-rescue support:** tasked alongside the RBDF Coast Guard over the maritime picture (§8.5).
- **Emergency comms relay:** a drone can act as an airborne relay to **temporarily restore connectivity** to a station or island whose VHF/cellular was destroyed, bridging it back to the mesh/satellite.
- **Station servicing:** inspection and sensor deployment for remote/relay sites that are hard to reach.
- **Built-in airspace deconfliction (key synergy):** because BACSWN already holds the real-time commercial + military air picture (§8.8), it can **safely deconflict its own UAS operations** against live traffic — a capability most standalone drone programs lack.
- **Integration & autonomy:** drones are tasked through the emergency/dispatch layer, stream telemetry/imagery over the mesh + satellite into the COP, and use edge-AI mission planning; all links are secured per §4.4 and flown under **BCAA UAS regulations**.

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10. Standards, Spectrum & Compliance

- **Spectrum:** licensed VHF mesh (148–174 MHz) requires allocation from **URCA**; application is a Month-1 critical-path item (risk R03).
- **Aviation:** advisory products conform to **ICAO/WMO SIGMET** format; local broadcast aligns to aviation VHF and ATIS-style conventions; coordination with **BCAA**.

- **Emissions:** per-flight CO_2 to **ICAO CORSIA** methodology (already implemented in-platform).
- **Security (§4.4): FIPS 140-3** crypto modules, **NIST SP 800-53**, **IEC 62443** (OT), signed firmware/OTA, hardware-rooted node identity, mutual TLS 1.3, post-quantum migration path.
- **Structural/survivability (§7): TIA-222-H / ASCE 7 Risk Category IV**, **ASTM E1996/E1886** missile-impact, site-specific surge elevation.

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11. Acceptance Criteria (per station)

A station is "commissioned" only when: (a) all three applicable transport layers pass a link-budget and live-traffic test; (b) the node sustains its rated extended-autonomy battery target (≥72 h standard, 96–120 h remote/relay) in a power-cut test; (c) it joins the mesh, exchanges `HEARTBEAT` / `OBSERVATION`, and participates in a consensus vote; (d) it runs local edge inference within latency budget; (e) it performs an autonomous-mode drill (hub disconnected) with continued local alerting; (f) an OTA A/B firmware update succeeds and rolls back cleanly; **(g) structural, missile-impact, and surge-elevation survivability are certified for Cat 4/5 design loads (§7);** and **(h) cryptographic provisioning is verified — secure-element key generation, certificate enrollment, mutual-TLS/signed-envelope traffic, signed-OTA verification, and tamper-zeroization (§4.4).**

Annex A — RF Link Budgets

A.1 Purpose & Method

This annex gives the RF link-budget basis for the VHF mesh and confirms the role of the satellite overlay. Method: **EIRP − path loss + RX gain − losses**, compared to receiver sensitivity with a fade margin, plus a **radio-horizon (line-of-sight) geometry check** — which, for VHF over sea, is the binding constraint.

A.2 VHF Mesh Parameters (baseline)

Parameter	Value
Band / mid-frequency	148–174 MHz / 162 MHz, narrowband
Radio	Motorola MTR3000, TX 50 W (47 dBm), configurable to 100 W
Antenna	6 dBi omni (mesh) / up to ~11 dBi Yagi (fixed long hops)
Feeder + connector loss	2 dB per end
Data rate	9.6 kbps
RX sensitivity	−112 dBm (typical, narrowband)
Target fade margin	≥ 20 dB (over-sea multipath / variability)
EIRP (omni)	**51 dBm**

A.3 Path Loss & Margin (assuming line-of-sight)

Hop	Free-space path loss	RX power	Margin over sensitivity
22 nm	108.8 dB	−53.8 dBm	58 dB
25 nm	109.9 dB	−54.9 dBm	57 dB
50 nm	116.0 dB	−61.0 dBm	51 dB
77 nm	119.7 dB	−64.7 dBm	47 dB
92 nm	121.3 dB	−66.3 dBm	46 dB

Finding: where line-of-sight exists, every mesh hop closes with **45–58 dB** of margin — far above the 20 dB target. **VHF links are not power-limited.**

A.4 The Binding Constraint — Radio Horizon

VHF is line-of-sight; over sea the limit is the radio horizon (4/3-earth): **d_{LOS} (nm) ≈ 2.22 ($\sqrt{h_1} + \sqrt{h_2}$)**, heights in metres.

Antenna height (each end)	LOS range
30 m	24 nm
50 m	31 nm

Antenna height (each end)	LOS range
100 m	44 nm
150 m	54 nm
200 m	63 nm

A 92 nm single hop would need **~430 m** of combined antenna/terrain height — impractical at the low-lying remote sites.

A.5 Design Consequences (validates §5–§7)

- **Single-hop VHF** is reliably engineered to **~30–45 nm** with practical 50–100 m masts (further where elevated terrain exists), with ample power margin.
- **Mid-range gaps** are bridged by **multi-hop routing** through intermediate stations — the routing graph (§6) is a *routing* topology, not a claim of direct LOS on every edge.
- **The remote SE edge (Inagua ↔ Mayaguana, 92 nm)** is **beyond practical VHF LOS**, confirming the architecture's decision that the **Iridium satellite overlay is their primary/critical path** (§5.3, §7.5) — not a fallback.
- Tropospheric ducting over warm sea can occasionally extend VHF far beyond horizon, but is **not relied upon** for design.

A.6 Satellite & Cellular

- **Iridium 9603 SBD** is a closed LEO system with its own managed link budget; the design relies on its global availability and hurricane-proof low-profile antenna, not a site-specific budget. Its low rate (~2.4 kbps) is sufficient for HEARTBEAT / ALERT / critical messaging.
- **Cellular** is opportunistic (coverage islands only), treated as bonus bandwidth, not budgeted.

A.7 Verification

Per-station SAT (§11; V&V plan BACSWN-VV-2026-006) includes a **measured link-budget + live-traffic test** per transport layer; field-measured margins are recorded in the RTM and must meet the ≥ 20 dB VHF target — or the hop is realized by relay/satellite.

Companion documents: BACSWN Master Framework Schedule (BACSWN-SCHED-2026-003); BACSWN Operations & Maintenance Plan (BACSWN-OM-2026-004); BACSWN Data Governance & Sovereignty Policy (BACSWN-GOV-2026-005); BACSWN Master V&V and ICD Plan (BACSWN-VV-2026-006); BACSWN Training & Capacity-Building Plan (BACSWN-TRN-2026-007); BACSWN SWOT & Risk Analysis (BACSWN-RISK-2026-008); Board Runbook (technology highlights); Distributed Intelligence Architecture Whitepaper (BACSWN-ARCH-2026-001).